Object Oriented Programming using Java

Assignment - 8

Collection Framework & Generics

Note: For all the program take user inputs to initialize the collection objects/arrays/strings and to perform the display operations of the collection objects always use loops.

1. Create an ArrayList to store integer numbers only. Take user input and store 10 numbers to the list. Perform each of the following operation and display the list.
   a. Display the list using for loop, for-each loop, and Iterator object.
   b. Update the 4th index value to 100.
   c. At the 6th index insert 500
   d. Remove the 2nd index value
   e. Remove 100 from the list
   f. Find the current index of 500.
   g. Sort the list in both ascending and descending order.
   h. Take an integer k as input and find the kth largest number if k is valid.
2. Write a java program to store 5 employee objects in an ArrayList. Each employee will have name, age and salary. Make sure the ArrayList object will store only employee objects. Take the employee details as input from the user and
   a. display the employee details.
   b. Find the average salary of all the employees and display it.
   c. Find the highest paid employee
   d. Sort the employee based on their name
   e. Remove the employees who are more than 60 years old.
3. You are given a sequence of operations to simulate using a Stack.
   a. Create a Stack<Integer>.
   b. Push 5 integers onto the stack.
   c. Pop an integer from the stack and display it.
   d. Check if a specific number is present in the stack.
   e. Peek the top element without removing it.
   f. Print the final stack after performing all operations.
4. Write a program to check if a given string containing parentheses is balanced. Use a stack to solve the problem.
5. Create a LinkedList to store 5 color names and perform the following operations. Display the linkedlist after each operation.
   a. Insert one more color in front.
   b. Insert a color in 3rd position.
   c. Replace the 2nd index color with another color name.
   d. Remove the color from the 4th index

   e. Display the first and last color name
   f. Remove the first and last colors.
6. You are tasked with managing a list of Product objects for an e-commerce application. Define a Product class with properties name, price and rating (double). Create a LinkedList<Product> to store 5 products, and add products to the list.
   a. Iterate over the list and display all product names, their prices, and rating.
   b. Find the product with the highest price (consider each product has a unique price) and display the product details.
   c. Sort the product based on their rating in descending order.
   d. Remove all the products with rating less than 2.5.
7. Write a Java program to create a priority queue, add some numbers and print out the elements of the priority queue.
   a. Display the head element of the queue.
   b. Remove the head element of the queue.
   c. Remove a number from the queue (not head).
   d. Check if the queue contains 10 or not.
8. You are designing a triage system for a hospital's emergency room, where patients are prioritized based on the severity of their condition. Define a Patient class with properties name and severity (integer, where a lower number means more severe). Create a PriorityQueue to store patients, ordered by severity. Add 5 patients to the queue. Display the patients in order of severity (highest severity first). Remove the most critical patient and display the remaining patients.
9. Write a Java program to create a HashMap to store student roll (key) and name (value).
   a. Display the student details.
   b. Add one more student to the collection.
   c. Extract the roll numbers from the map and store into another collection, then display.
   d. Extract the student names from map and display.
   e. Search a student by name.
   f. Update name of a student.
   g. Remove a student by using his roll number.
10. Write a program to find the frequency of elements in an array. Use a Map to store and display the frequencies.
11. You are given a list of Student names.
   a. Create a HashSet<String> to store the student names.
   b. Add 5 students to the set.
   c. Check if a particular student is already in the set.
   d. Iterate over the set and print all students' names.
   e. Remove a student from the set.
   f. Print the final list of students in the set.
12. Write a java program to remove duplicates from an array using a HashSet.